Chapter 3. Connectedness and Compactness

Connected and path-connected spaces, components, compact and limit point compact spaces, local compactness

Contents

3.1 Connected spaces **2**

3.2 Connected subspaces of the real line **3**

3.3 Components and local connectedness **4**

3.4 Compact spaces **5**

3.5 Compact subspaces of the real line **6**

3.6 Limit point compactness **8**

3.7 Local compactness **8**

3.1 Connected spaces

Motivation. The intermediate value theorem is really a statement about the shape of $[a, b]$: it cannot be split into two disjoint open pieces. Connectedness abstracts this, and continuous images of connected spaces are connected, which gives the intermediate value theorem in any setting.

A **separation** of X is a pair U, V of disjoint nonempty open sets with $X = U \cup V$. X is **connected** if no separation exists; equivalently, the only sets both open and closed are $\emptyset$ and X.

Lemma 3.1 (23.1)

If $Y \subset X$, a separation of Y is a pair of disjoint nonempty sets A, B with $A \cup B = Y$, neither containing a limit point of the other.

Proof. If A, B separate Y, A is closed in Y, so $A = \overline{A} \cap Y$ (Theorem 2.14) and $\overline{A} \cap B = \emptyset$: B contains no limit point of A; symmetrically. Conversely, if $\overline{A} \cap B = \emptyset = A \cap \overline{B}$, then $\overline{A} \cap Y = A$ and $\overline{B} \cap Y = B$ are closed in Y, so their complements B, A are open in Y. ■

Examples: the indiscrete topology is connected; $\mathbb{Q}$ is totally disconnected (its only connected subspaces are points); $Y = [-1, 0) \cup (0, 1]$ is disconnected.

Lemma 3.2 (23.2)

If C, D separate X and $Y \subset X$ is connected, then $Y \subset C$ or $Y \subset D$.

Proof. $C \cap Y$ and $D \cap Y$ are open in Y and disjoint with union Y; if both were nonempty they would separate Y. ■

Theorem 3.3 (23.3)

The union of a collection of connected subspaces of X with a point in common is connected.

Proof. Let p be the common point and suppose C, D separate $Y = \bigcup A_\alpha$, with $p \in C$. Each A_α lies in C or D (Lemma 3.2), hence in C as $p \in A_\alpha$. So $D = \emptyset$, a contradiction. ■

Theorem 3.4 (23.4)

If A is connected and $A \subset B \subset \overline{A}$, then B is connected.

Proof. If C, D separate B, then $A \subset C$, say (Lemma 3.2); then $\overline{A} \subset \overline{C}$, and $\overline{C} \cap D = \emptyset$ (Lemma 3.1), so $B \cap D \subset \overline{A} \cap D = \emptyset$. ■

Theorem 3.5 (23.5)

The image of a connected space under a continuous map is connected.

Proof. If A, B separate $f(X)$, then $f^{-1}(A), f^{-1}(B)$ separate X. ■

Theorem 3.6 (23.6)

A finite cartesian product of connected spaces is connected.

Proof. For $X \times Y$ fix $a \times b$. For each x, $T_x = (X \times b) \cup (x \times Y)$ is connected (Theorem 3.3, with common point $x \times b$). All T_x contain $a \times b$, and their union is $X \times Y$. Induct for more factors. ■

Arbitrary products are connected in the product topology (the set of points differing from a fixed point in finitely many coordinates is a connected dense subset), but $\mathbb{R}^\omega$ in the box topology is not: bounded and unbounded sequences form a separation.

3.2 Connected subspaces of the real line

Motivation. The real line is connected because of the least upper bound property; from this we get that intervals are connected and, through path connectedness, that $\mathbb{R}^n$ and spheres are connected.

A simply ordered set L with more than one element is a **linear continuum** if it has the least upper bound property and for $x < y$ there is z with $x < z < y$.

Theorem 3.7 (24.1)

A linear continuum L in the order topology is connected, and so are its intervals and rays.

Proof. It suffices to show convex subsets Y are connected. Suppose A, B separate Y, $a \in A$, $b \in B$, $a < b$. $[a, b] \subset Y$, so $A_0 = A \cap [a, b]$, $B_0 = B \cap [a, b]$ separate $[a, b]$. Let $c = \sup A_0$. If $c \in B_0$, then $c \neq a$, so $c > a$; B_0 is open in $[a, b]$, so some $(d, c] \subset B_0$, and then d is a smaller upper bound for A_0 (since $(d, c]$ misses A_0 and c is the sup): contradiction. If $c \in A_0$, then $c < b$ and some $[c, e) \subset A_0$; choosing z with $c < z < e$ gives $z \in A_0$, contradicting $c = \sup A_0$. ■

Corollary 3.8 (24.2)

$\mathbb{R}$ is connected, and so are its intervals and rays.

Theorem 3.9 (24.3, intermediate value theorem)

Let $f : X \to Y$ be continuous, X connected, Y ordered in the order topology. If $a, b \in X$ and r lies between $f(a)$ and $f(b)$, then $f(c) = r$ for some $c \in X$.

Proof. Otherwise $f(X) \cap (-\infty, r)$ and $f(X) \cap (r, +\infty)$ separate the connected set $f(X)$. ■

Other linear continua: $I \times I$ in the dictionary order (the ordered square), and the **long line** $L = S_\Omega \times [0, 1)$. A **path** from x to y is a continuous $f : [a, b] \to X$ with $f(a) = x$, $f(b) = y$; X is **path connected** if every two points are joined by a path. Path connected spaces are connected (Theorems 3.3, 3.5, 3.8).

Example 3.1

(a) The unit ball B^n and punctured space $\mathbb{R}^n - \{0\}$ $(n > 1)$ are path connected, hence so is S^{n-1} (its image under $x \mapsto x/\|x\|$).
(b) The ordered square is connected but not path connected: a path from the least to the greatest element would be surjective, and the uncountably many disjoint open intervals $U_x = (x \times 0, x \times 1)$ would have disjoint open preimages each containing a rational.
(c) The **topologist's sine curve** $\overline{S}$, the closure in $\mathbb{R}^2$ of $S = \{(x, \sin(1/x)) : 0 < x \leq 1\}$, is connected (Theorem 3.4) but not path connected: if $f : [a, c] \to \overline{S}$ were a path from 0×0 into S, let $b = \sup\{t : f(t) \in 0 \times [-1, 1]\}$; on $(b, c]$, $f(t) \in S$, and choosing $t_n \downarrow b$ with $\sin(1/x(t_n)) = (-1)^n$ contradicts the continuity of the second coordinate at b.

3.3 Components and local connectedness

Motivation. A disconnected space breaks into maximal connected pieces, its components. Counting components is the simplest topological invariant, used for instance to show that $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic.

Define $x \sim y$ if some connected subspace contains both; the classes are the **components** of X.

Theorem 3.10 (25.1)

The components of X are connected disjoint subspaces whose union is X, and each nonempty connected subspace meets only one of them.

Proof. $\sim$ is an equivalence relation (transitivity by Theorem 3.3). A connected A meeting components C_1, C_2 in x_1, x_2 gives $x_1 \sim x_2$. For connectedness of a component C, fix $x_0 \in C$; each $x \in C$ lies in a connected $A_x \ni x_0$, contained in C by the previous sentence, and $C = \bigcup A_x$ is connected by Theorem 3.3. ■

Similarly, define $x \sim y$ if a path joins them; the classes are the **path components**.

Theorem 3.11 (25.2)

The path components of X are path-connected disjoint subspaces with union X, and each nonempty path-connected subspace meets only one of them.

Proof. Transitivity: concatenate paths $f : [a, b] \to X$ and $g : [b, c] \to X$ via the pasting lemma. ■

Components are closed (the closure of a connected set is connected); path components need not be (the topologist's sine curve has two path components, S and $0 \times [-1, 1]$, the first not closed). X is **locally connected at** x if every neighbourhood of x contains a connected neighbourhood of x; similarly **locally path connected**. Intervals in $\mathbb{R}$ are both; $\mathbb{Q}$ is neither; the topologist's sine curve is connected but not locally connected.

Theorem 3.12 (25.3)

X is locally connected iff for every open U each component of U is open in X.

Proof. ($\Rightarrow$) If C is a component of U and $x \in C$, a connected neighbourhood $V \subset U$ of x lies in C.

($\Leftarrow$) For a neighbourhood U of x, the component of U containing x is a connected neighbourhood inside U. ■

Theorem 3.13 (25.4)

X is locally path connected iff for every open U each path component of U is open in X.

Theorem 3.14 (25.5)

Each path component of X lies in a component. If X is locally path connected, components and path components coincide.

Proof. The first claim since path connected sets are connected. Let C be a component and $P \subset C$ a path component. The path components of X are open (Theorem 3.13 with $U = X$); Q, the union of the other path components meeting C, is open, so P, Q would separate C unless $Q = \emptyset$. Hence $P = C$. ■

3.4 Compact spaces

Motivation. On $[a,b]$ continuous functions are bounded and attain their extrema, and they are uniformly continuous. The property responsible is compactness: every open cover has a finite subcover. It lets us pass from local information to global information, and it is preserved by continuous maps and products.

A collection of subsets **covers** X if its union is X; it is an **open covering** if its members are open. X is **compact** if every open covering contains a finite subcollection covering X. Examples: $\mathbb{R}$ is not compact; $\{0\}\cup\{1/n\}$ is compact; finite spaces and spaces with the finite complement topology are compact.

Lemma 3.15 (26.1)

A subspace $Y \subset X$ is compact iff every covering of Y by sets open in X contains a finite subcollection covering Y.

Proof. Open sets of Y are intersections $Y \cap A$ with A open in X; translate coverings in both directions. ■

Theorem 3.16 (26.2)

Every closed subspace of a compact space is compact.

Proof. Given a covering $\mathcal{A}$ of a closed Y by sets open in X, $\mathcal{A}\cup\{X-Y\}$ covers X; a finite subcover, minus $X-Y$, covers Y. ■

Theorem 3.17 (26.3)

Every compact subspace of a Hausdorff space is closed.

Lemma 3.18 (26.4)

If Y is a compact subspace of a Hausdorff space X and $x_0 \notin Y$, there are disjoint open $U \ni x_0$ and $V \supset Y$.

Proof. For each $y \in Y$ choose disjoint neighbourhoods $U_y \ni x_0$, $V_y \ni y$. Finitely many $V_{y_1},\dots,V_{y_n}$ cover Y; take $V = \bigcup V_{y_i}$ and $U = \bigcap U_{y_i}$. This proves Theorem 3.17 too: $X-Y$ is open. ■

Theorem 3.19 (26.5)

The image of a compact space under a continuous map is compact.

Proof. If $\mathcal{A}$ covers $f(X)$ by open sets of Y, $\{f^{-1}(A)\}$ covers X; finitely many suffice. ■

Theorem 3.20 (26.6)

A continuous bijection $f : X \to Y$ from a compact space to a Hausdorff space is a homeomorphism.

Proof. For closed $A \subset X$: A is compact (Theorem 3.16), $f(A)$ is compact (3.19), hence closed (3.17). So f^{-1} is continuous. ■

Lemma 3.21 (26.8, the tube lemma)

If Y is compact and N is an open set of $X \times Y$ containing the slice $x_0 \times Y$, then N contains a tube $W \times Y$ with W a neighbourhood of x_0.

Proof. Cover $x_0 \times Y$ by basis elements $U \times V \subset N$ meeting it; it is compact (homeomorphic to Y), so finitely many $U_i \times V_i$ cover it. Let $W = \bigcap U_i$. Every $x \times y$ with $x \in W$ lies in some $U_i \times V_i$ (choose i with $y \in V_i$), so $W \times Y \subset N$. ■

Theorem 3.22 (26.7)

A product of finitely many compact spaces is compact.

Proof. Let $\mathcal{A}$ be an open cover of $X \times Y$. For each x_0, finitely many members cover $x_0 \times Y$; their union N contains a tube $W_{x_0} \times Y$. Finitely many W_{x_i} cover X, and the corresponding finitely many members of $\mathcal{A}$ cover $X \times Y$. ■

A collection $\mathcal{C}$ has the **finite intersection property** if every finite subcollection has nonempty intersection.

Theorem 3.23 (26.9)

X is compact iff every collection $\mathcal{C}$ of closed sets with the finite intersection property has $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$.

Proof. Pass to complements: $\{X - C\}$ is an open cover iff $\bigcap C = \emptyset$, and a finite subcollection covers iff the corresponding finite intersection is empty. ■

In particular nested sequences $C_1 \supset C_2 \supset \cdots$ of nonempty closed sets in a compact space have nonempty intersection.

3.5 Compact subspaces of the real line

Motivation. We now prove that $[a, b]$ really is compact, and hence that closed bounded sets in $\mathbb{R}^n$ are compact. The extreme value theorem, the Lebesgue number lemma and uniform continuity all follow.

Theorem 3.24 (27.1)

If X is simply ordered with the least upper bound property, every closed interval $[a, b]$ in X is compact (order topology).

Proof. Let $\mathcal{A}$ be a covering of $[a, b]$ by sets open in $[a, b]$.

Step 1. for $x \in [a, b)$ there is $y > x$ such that $[x, y]$ is covered by at most two members of $\mathcal{A}$ (if x has an immediate successor y, use the members containing x, y; otherwise take $A \ni x$ containing $[x, c)$ and any $y \in (x, c)$).

Step 2. let C be the set of $y > a$ in $[a, b]$ such that $[a, y]$ is finitely covered; $C \neq \emptyset$ by Step 1. Let $c = \sup C$.

Step 3. $c \in C$: choose $A \ni c$ containing some $(d, c]$; if $c \notin C$, some $z \in C \cap (d, c)$, and $[a, z]$ plus A covers $[a, c]$.

Step 4. $c = b$: otherwise Step 1 extends past c. ■

Corollary 3.25 (27.2)

Every closed interval in $\mathbb{R}$ is compact.

Theorem 3.26 (27.3)

$A \subset \mathbb{R}^n$ is compact iff it is closed and bounded (in the euclidean or square metric).

Proof. If A is compact, it is closed (Theorem 3.17) and covered by finitely many open balls $B_\rho(0, m)$, hence bounded. Conversely, a closed bounded A lies in a cube $[-P, P]^n$, compact by Theorems 3.22 and 3.25; apply Theorem 3.16. ■

Theorem 3.27 (27.4, extreme value theorem)

If $f : X \to Y$ is continuous, X compact and Y ordered, there are c, d with $f(c) \leq f(x) \leq f(d)$ for all x.

Proof. $A = f(X)$ is compact. If A had no largest element, the rays $(-\infty, a)$, $a \in A$, would cover A; a finite subcover $(-\infty, a_i)$ misses the largest a_i. So A has a largest element $f(d)$; similarly a smallest. ■

For $A \subset X$ in a metric space, $d(x, A) = \inf_{a \in A} d(x, a)$ is continuous in x (indeed $|d(x, A) - d(y, A)| \leq d(x, y)$).

Lemma 3.28 (27.5, Lebesgue number lemma)

Let $\mathcal{A}$ be an open covering of a compact metric space X. There is $\delta > 0$ such that every subset of diameter less than δ lies in some member of $\mathcal{A}$.

Proof. If $X \in \mathcal{A}$ take any δ. Otherwise choose a finite subcover $A_1, \ldots, A_n$, let $C_i = X - A_i$ and $f(x) = \frac{1}{n} \sum d(x, C_i)$. $f > 0$ (each x lies in some A_i, so $d(x, C_i) > 0$) and continuous, so $f \geq \delta > 0$ (Theorem 3.27). If $\operatorname{diam} B < \delta$ and $x_0 \in B$, then $B \subset B(x_0, \delta)$; choosing m with $d(x_0, C_m)$ maximal gives $d(x_0, C_m) \geq f(x_0) \geq \delta$, so $B(x_0, \delta) \subset A_m$. ■

Theorem 3.29 (27.6, uniform continuity theorem)

A continuous map from a compact metric space to a metric space is uniformly continuous.

Proof. Given ϵ, cover X by the open sets $f^{-1}(B(y, \epsilon/2))$ and let δ be a Lebesgue number. If $d(x_1, x_2) < \delta$, $\{x_1, x_2\}$ lies in one of these sets, so $d(f(x_1), f(x_2)) < \epsilon$. ■

A point x is **isolated** if $\{x\}$ is open.

Theorem 3.30 (27.7)

A nonempty compact Hausdorff space without isolated points is uncountable.

Proof.

Step 1. for any nonempty open U and point x, there is a nonempty open $V \subset U$ with $x \notin \overline{V}$ (choose $y \in U$, $y \neq x$, which exists since $U \neq \{x\}$; separate x and y by disjoint $W_1 \ni x$, $W_2 \ni y$; $V = W_2 \cap U$).

Step 2. given $f : \mathbb{Z}_+ \to X$, apply Step 1 recursively to get nonempty open $V_1 \supset V_2 \supset \cdots$ with $f(n) \notin \overline{V_n}$ and $\overline{V_{n+1}} \subset V_n$. The nested closed sets $\overline{V_n}$ have a common point x (Theorem 3.23), and $x \neq f(n)$ for all n. ■

Corollary 3.31 (27.8)

Every closed interval in $\mathbb{R}$ is uncountable.

3.6 Limit point compactness

Motivation. In analysis compactness is often defined with sequences (Bolzano–Weierstrass). We compare the cover, limit-point and sequential versions and show that they agree for metric spaces.

X is **limit point compact** if every infinite subset has a limit point, and **sequentially compact** if every sequence has a convergent subsequence.

Theorem 3.32 (28.1)

Compactness implies limit point compactness, but not conversely.

Proof. If A has no limit point, A is closed and hence compact, and each $a \in A$ has a neighbourhood U_a with $U_a \cap A = \{a\}$. The U_a cover the compact A, finitely many suffice, and each contains one point of A: A is finite. Counterexample to the converse: $X = \mathbb{Z}_+ \times Y$ with Y a two-point indiscrete space (every point has a limit point), or S_Ω below. ■

Theorem 3.33 (28.2)

For a metrizable space X the following are equivalent:
(1) compact;
(2) limit point compact;
(3) sequentially compact.

Proof. (1)⇒(2) is Theorem 3.32. (2)⇒(3): if the range of (x_n) is finite, a constant subsequence converges; otherwise it has a limit point x, and choosing $x_{n_k} \in B(x, 1/k)$ with n_k increasing (possible since each ball contains infinitely many points of the range, Theorem 2.19) gives a convergent subsequence. (3)⇒(1): first, the Lebesgue number lemma holds: if not, there are C_n of diameter $< 1/n$ in no member of $\mathcal{A}$; pick $x_n \in C_n$ and a convergent subsequence $x_{n_i} \to a$; $a \in A \in \mathcal{A}$ with $B(a, \epsilon) \subset A$, and for large i, $C_{n_i} \subset B(a, \epsilon)$, a contradiction. Second, X is **totally bounded**: if some ϵ admits no finite cover by ϵ-balls, choose $x_{n+1} \notin \bigcup_{i \le n} B(x_i, \epsilon)$; the sequence has no convergent subsequence since all distances are $\ge \epsilon$. Finally, with δ a Lebesgue number, cover X by finitely many $\delta/3$-balls; each lies in a member of $\mathcal{A}$. ■

Example 3.2

S_Ω is limit point compact but not compact: an infinite subset contains a countably infinite subset B, which has an upper bound b (Theorem 1.27), so $B \subset [a_0, b]$, which is compact (Theorem 3.24); hence B has a limit point. S_Ω is not compact since the sections S_α (or the open intervals $[a_0, \alpha)$) cover it without a finite subcover. It is also sequentially compact.

3.7 Local compactness

Motivation. $\mathbb{R}^n$ is not compact, but it is compact near every point, and adding one point at infinity makes it compact ($\mathbb{R}^2 \cup \{\infty\}$ is the sphere). Local compactness is exactly the condition for this one-point compactification to be Hausdorff.

X is **locally compact at** x if some compact C contains a neighbourhood of x, and locally compact if this holds at every point. $\mathbb{R}$, $\mathbb{R}^n$, every simply ordered set with the least upper bound property are locally compact; $\mathbb{Q}$ and $\mathbb{R}^\omega$ are not.

Theorem 3.34 (29.1)

X is locally compact Hausdorff iff there is a space Y with:
(1) X is a subspace of Y;
(2) $Y - X$ is a single point;
(3) Y is compact Hausdorff. Two such Y are homeomorphic by a homeomorphism fixing X.

Proof. Uniqueness: map $Y \to Y'$ by the identity on X and $p \mapsto q$; open sets of Y containing p are complements of compact subsets of X ($Y - U$ is closed in Y, hence compact, and lies in X), and the same holds in Y'.

Existence: let $Y = X \cup \{\infty\}$ with open sets the open sets of X and the sets $Y - C$ for compact $C \subset X$. This is a topology (uses that compact sets in the Hausdorff X are closed); X is a subspace; Y is compact (a cover contains some $Y - C$, and C needs finitely many more); Y is Hausdorff (separate x from ∞ by $\operatorname{Int} C \ni x$ and $Y - C$, where C is a compact set containing a neighbourhood of x). Conversely, a subspace of a compact Hausdorff space with one point removed is Hausdorff and locally compact (separate x from the removed point by disjoint open $U \ni x$, $V \ni \infty$; $C = Y - V$ is compact and contains U). ■

Y is the **one-point compactification**; e.g. $\mathbb{R} \cup \{\infty\} \cong S^1$ and $\mathbb{R}^n \cup \{\infty\} \cong S^n$ (stereographic projection).

Theorem 3.35 (29.2)

A Hausdorff space X is locally compact iff for every x and neighbourhood U of x there is a neighbourhood V of x with $\overline{V}$ compact and $\overline{V} \subset U$.

Proof. ($\Leftarrow$) Take $C = \overline{V}$.

($\Rightarrow$) In the one-point compactification Y, $C = Y - U$ is closed, hence compact; by Lemma 3.18 there are disjoint open $V \ni x$, $W \supset C$. Then $\overline{V}$ (closure in Y) misses W, so $\overline{V} \subset U$; and $\overline{V}$ is compact as a closed subset of Y. ■

Corollary 3.36 (29.3)

A closed or open subspace of a locally compact Hausdorff space is locally compact.

Proof. Closed A: if C is compact containing a neighbourhood U of x, then $C \cap A$ is compact and contains $U \cap A$. Open A: by Theorem 3.35, $x \in V \subset \overline{V} \subset A$ with $\overline{V}$ compact. ■

Corollary 3.37 (29.4)

X is homeomorphic to an open subspace of a compact Hausdorff space iff X is locally compact Hausdorff.